

**MBA
USP
ESALQ**

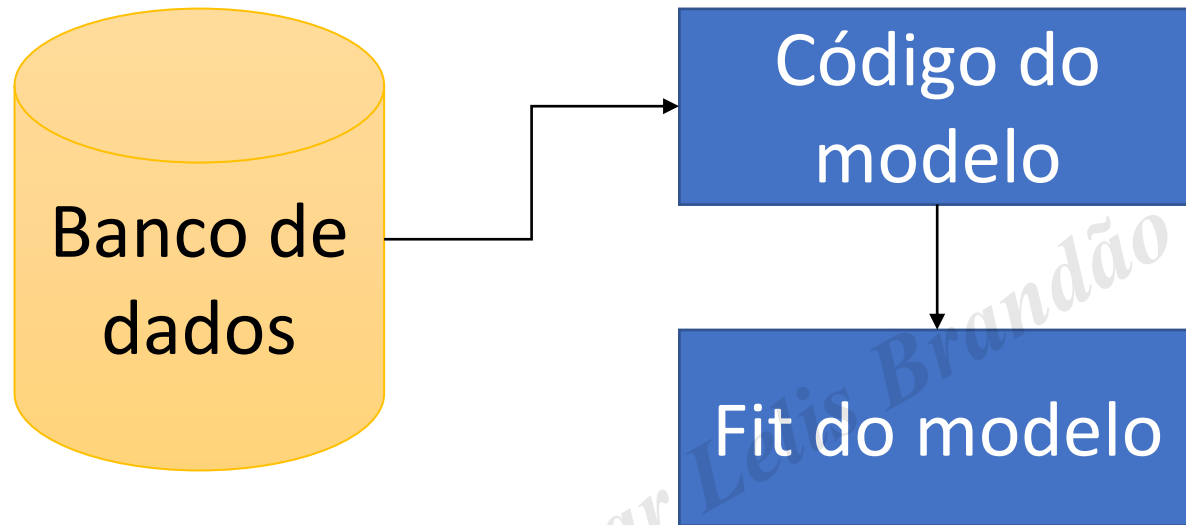
**BIG DATA E DEPLOYMENT DE
MODELOS**

Helder Prado Santos

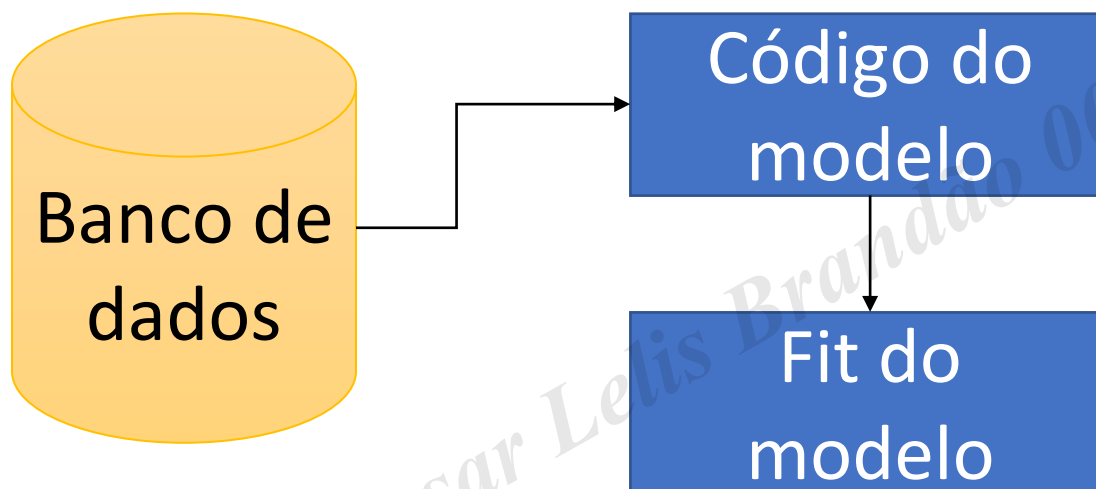
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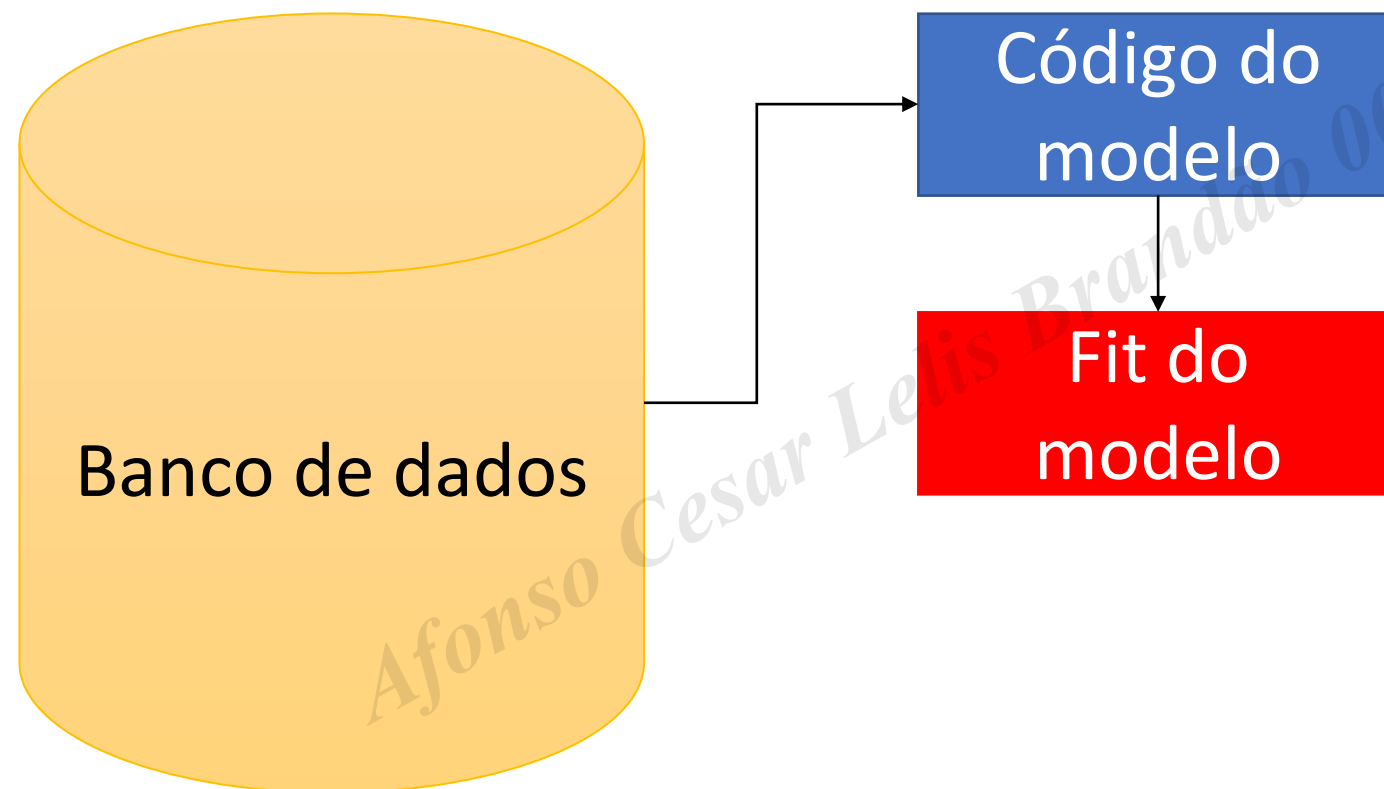
Vamos desenvolver um modelo



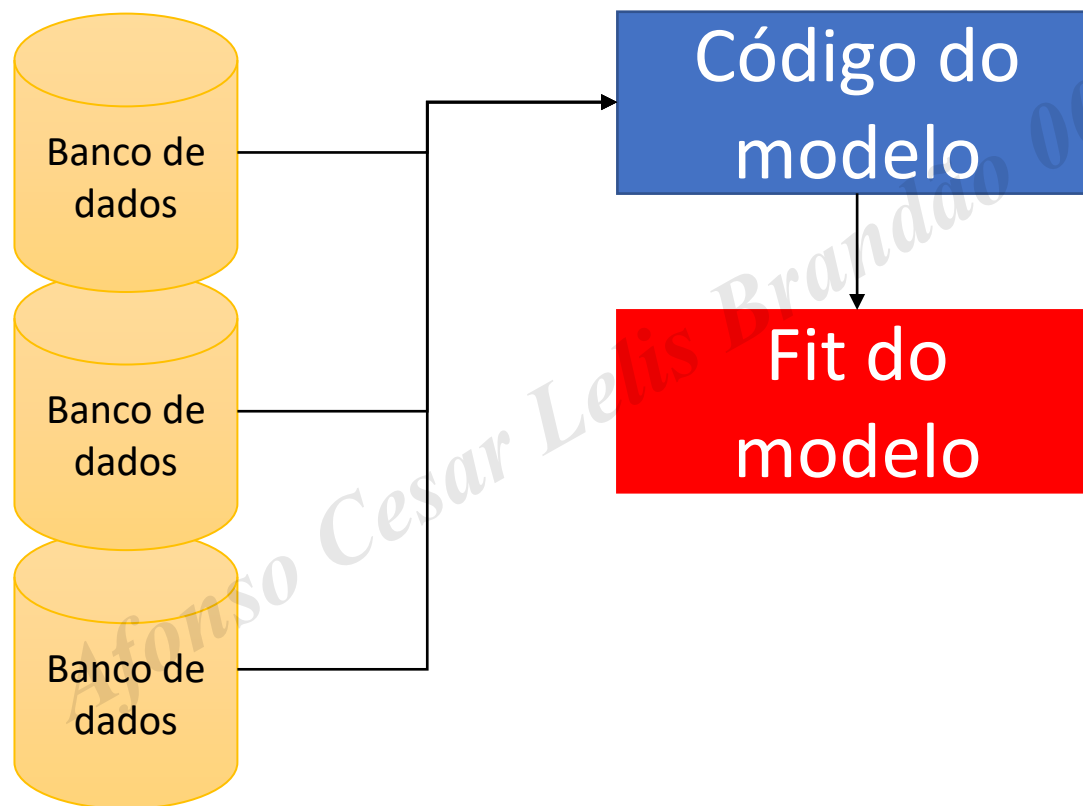
E se?



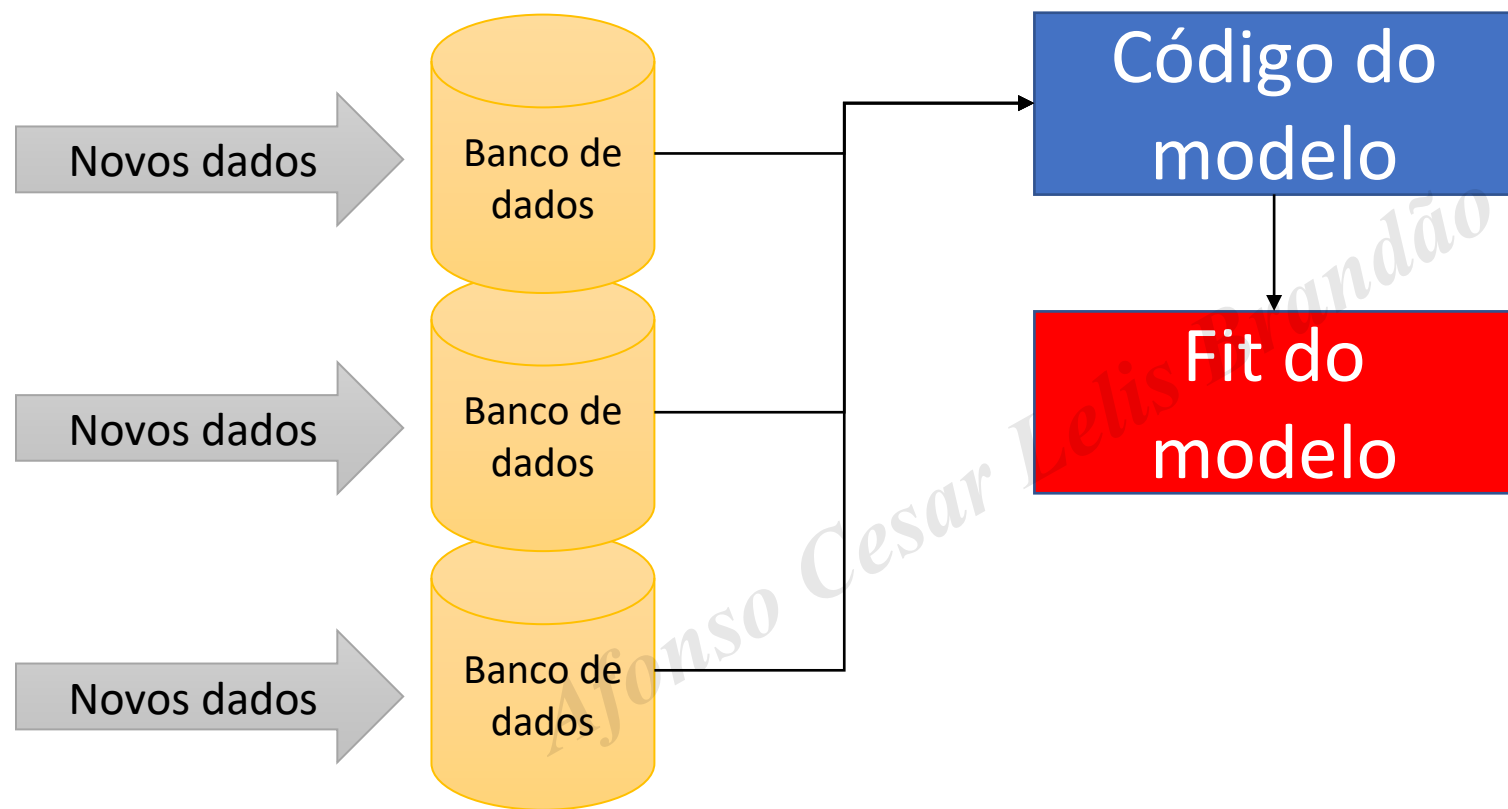
E se?



E se?



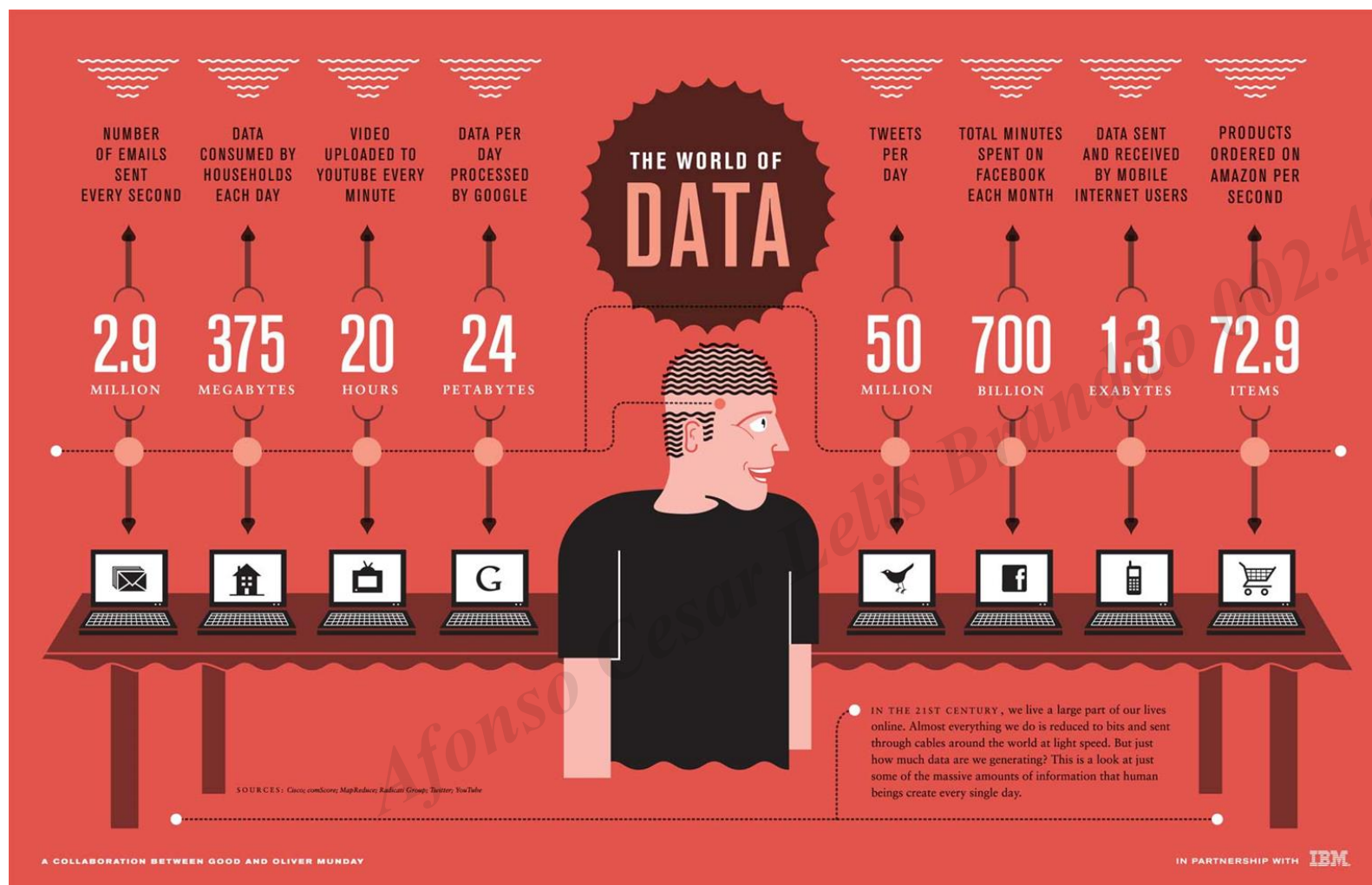
E se?



Alguns problemas encontrados...

1. E se nosso banco de dados crescer?
2. E se tivermos vários bancos de dados?
3. E se novos dados chegarem ao longo do tempo?

O nosso mundo de dados



Fonte: <https://hexanika.com/history-of-biq-data/>

Entrando no mundo do Big Data

- É uma ferramenta?
- É só ter grandes volumes de dados?

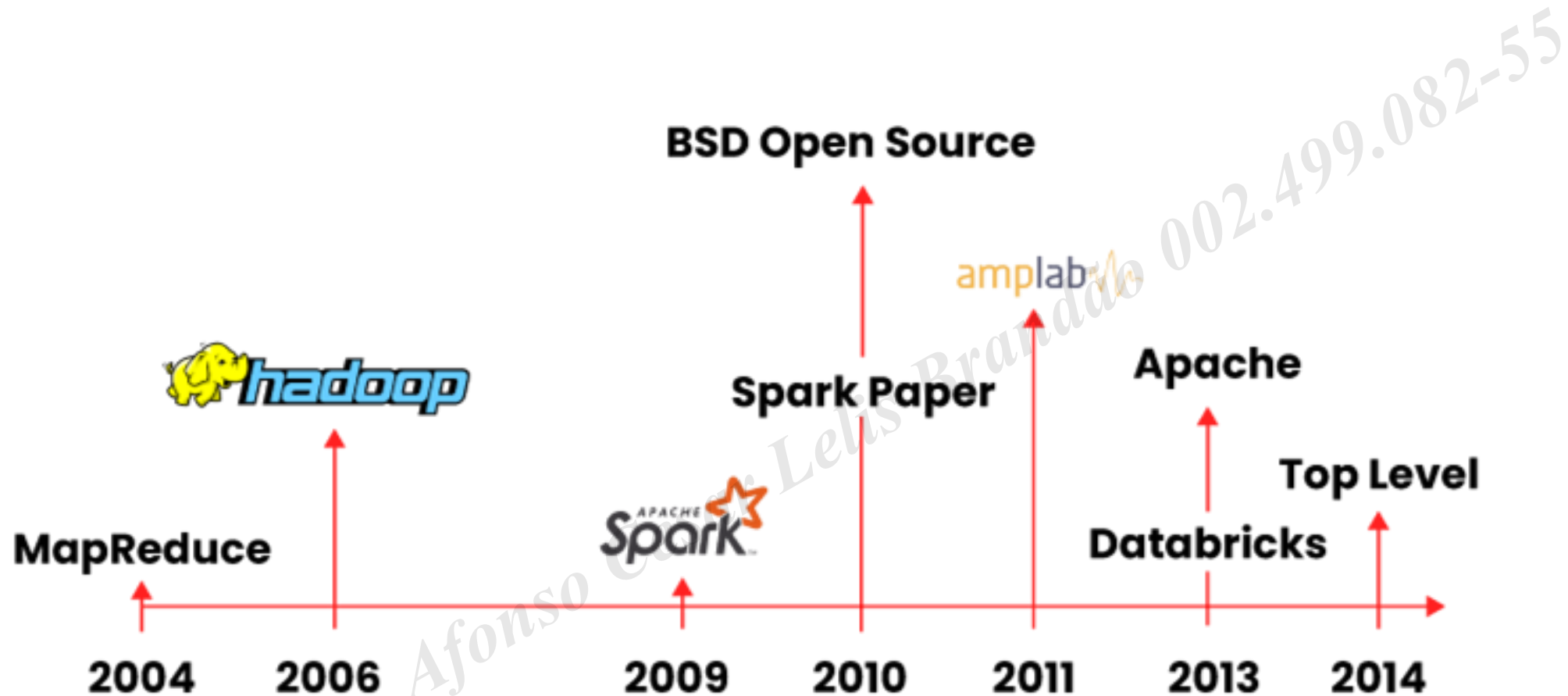
Afonso Cesar Lelis Brandão 002.499.082-55

Big Data: Os 5V



Fonte: <https://www.cortex-intelligence.com/blog/os-5-vs-do-big-data>

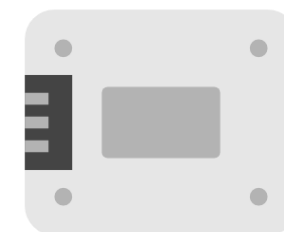
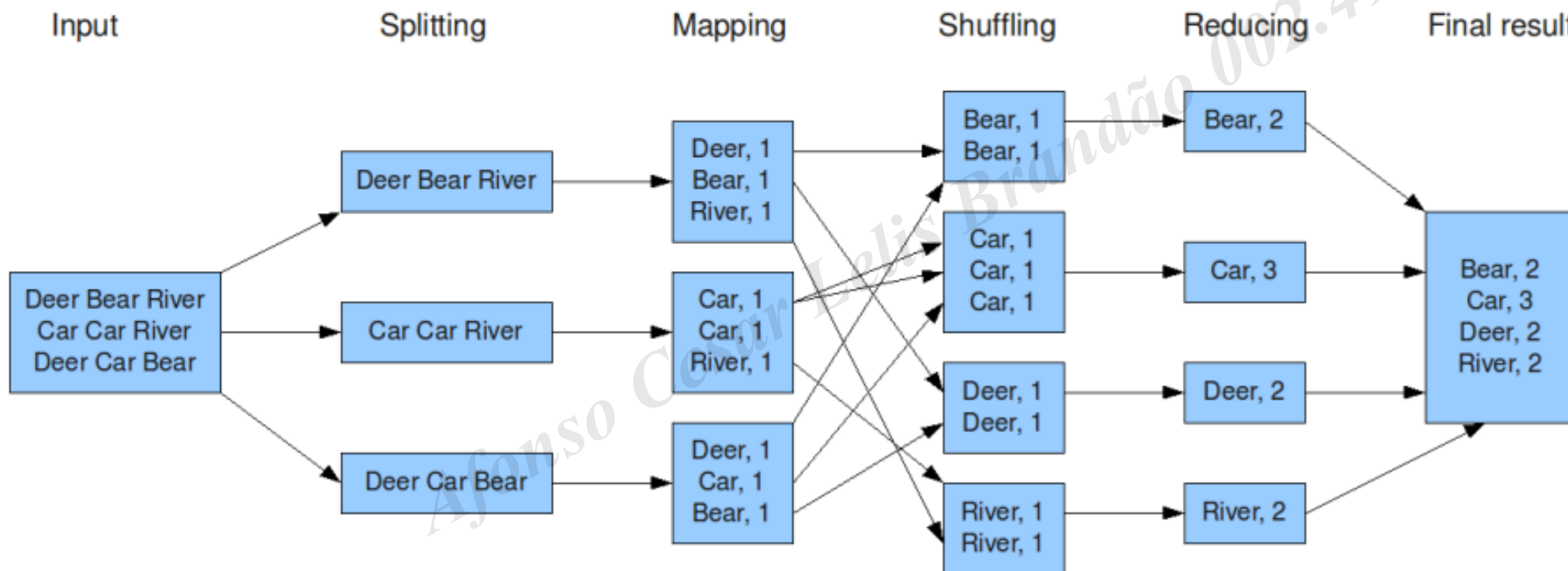
Sua história e evolução



Fonte: <https://betravingknows.com/weekly-reports/data-analytics/2018/05/how-big-data-shaped-todays-casino-and-why-you-should-care/>

Hadoop e a introdução do MapReduce

The overall MapReduce word count process

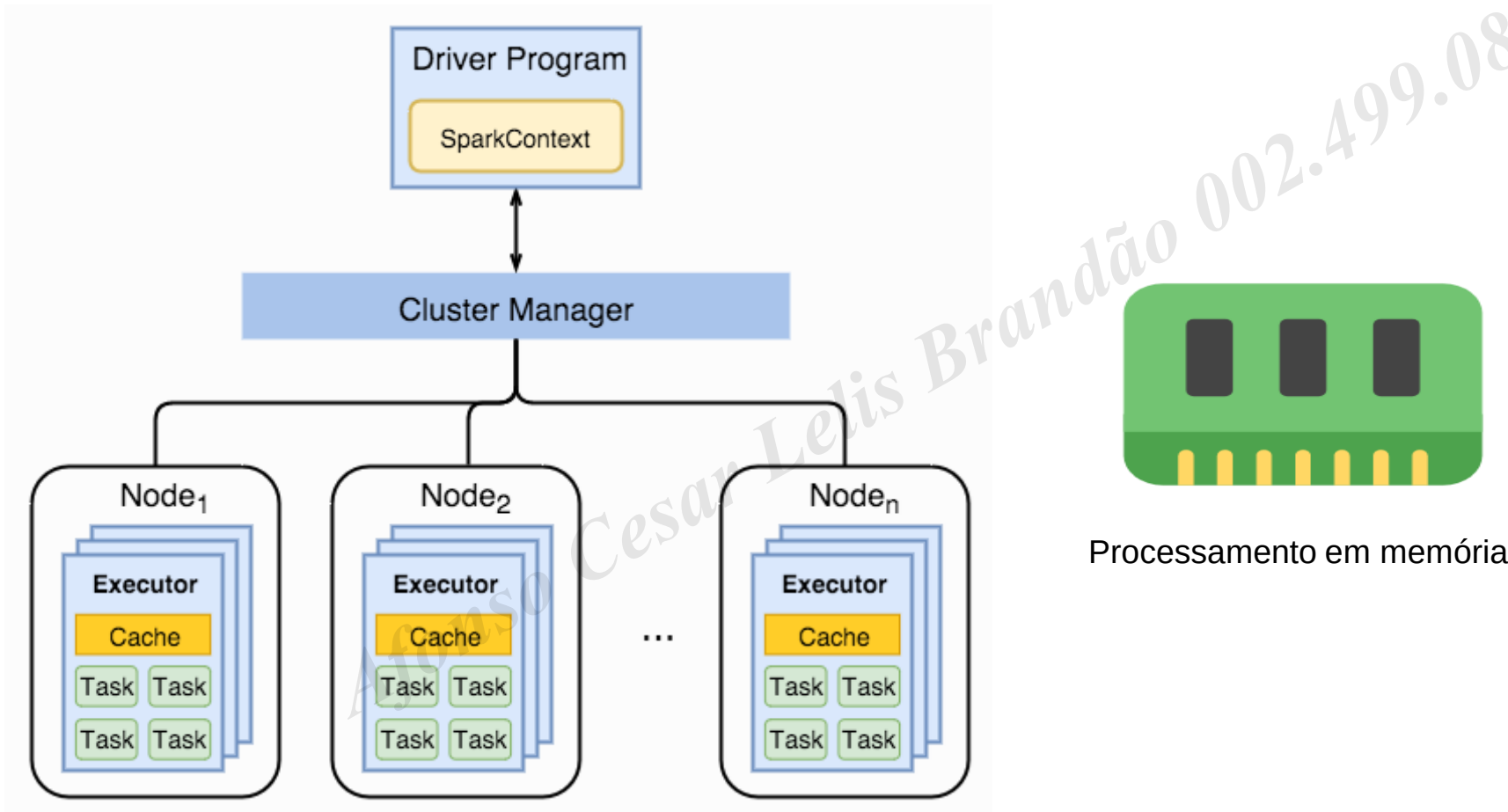


Processamento em disco

Fonte: <https://betravingknows.com/weekly-reports/data-analytics/2018/05/how-big-data-shaped-todays-casino-and-why-you-should-care/>

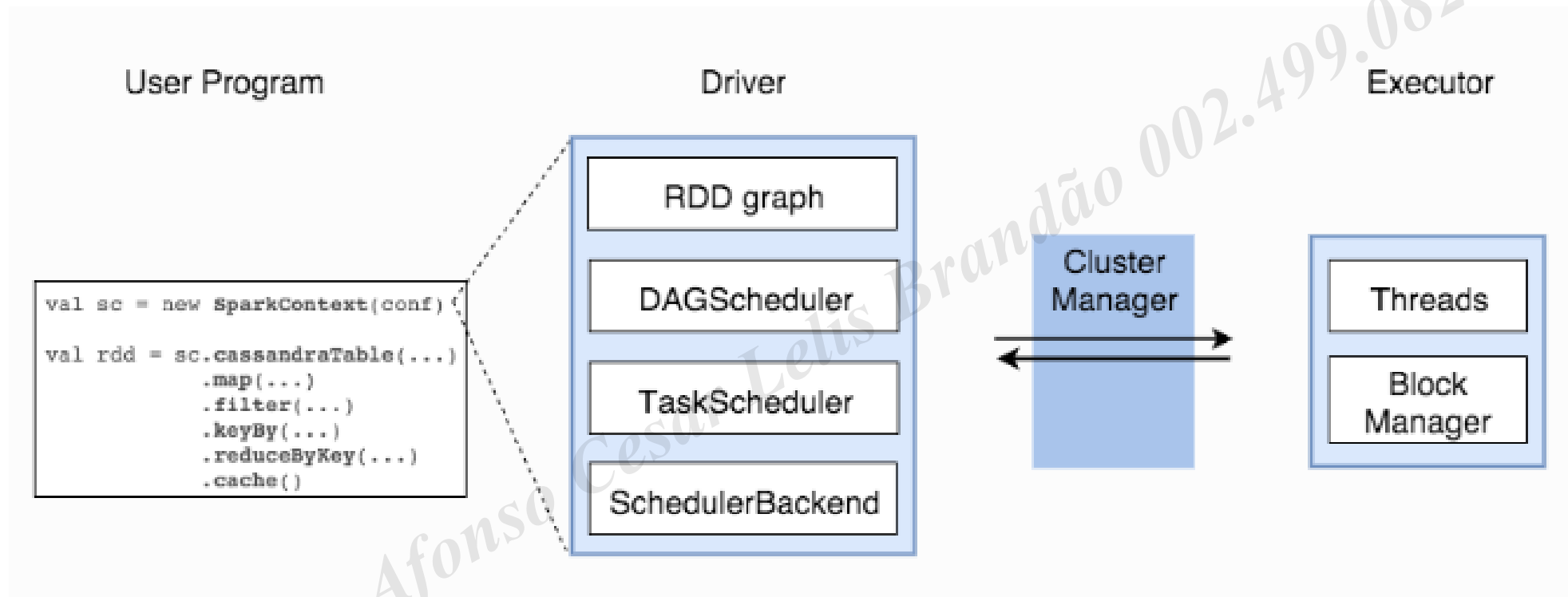
Spark e a introdução do RDD

Resilient distributed datasets



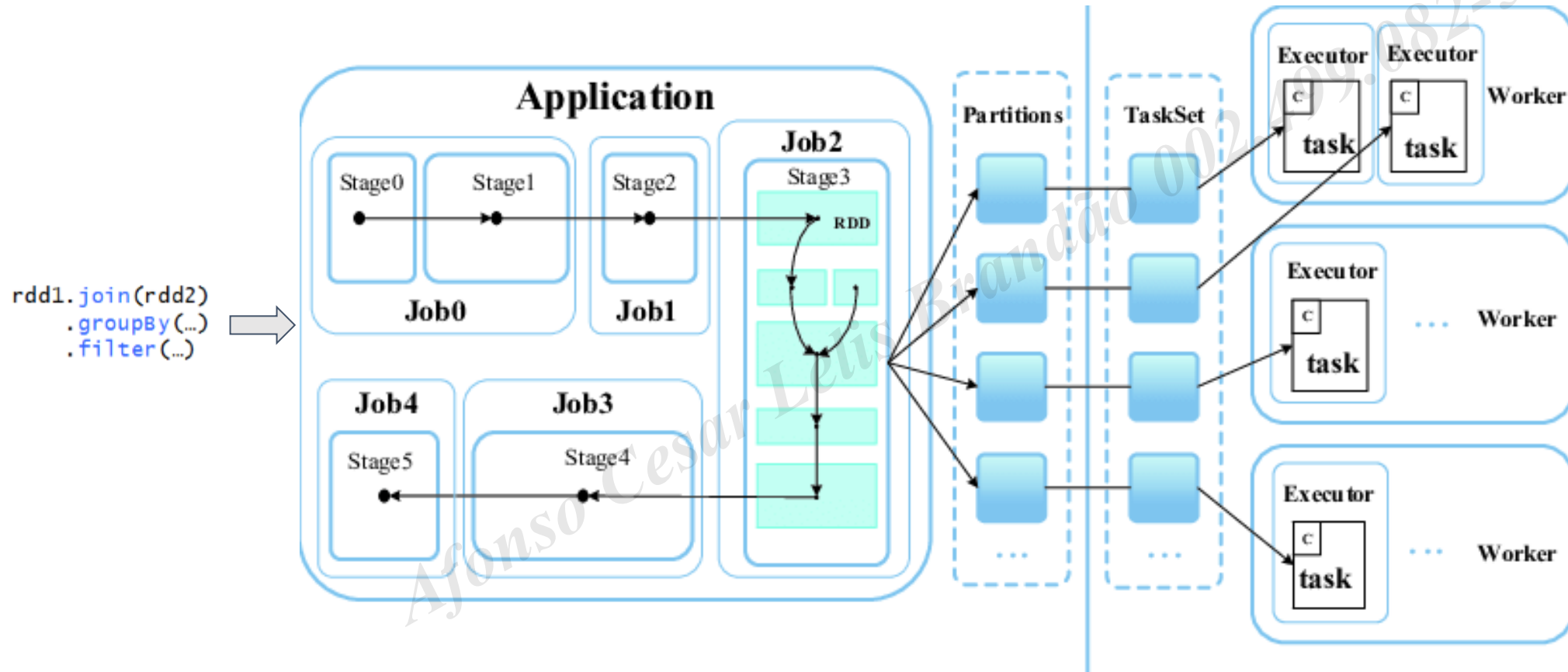
Fonte: <https://runawayhorse001.github.io/LearningApacheSpark/introduction.html>

Spark e a introdução do RDD



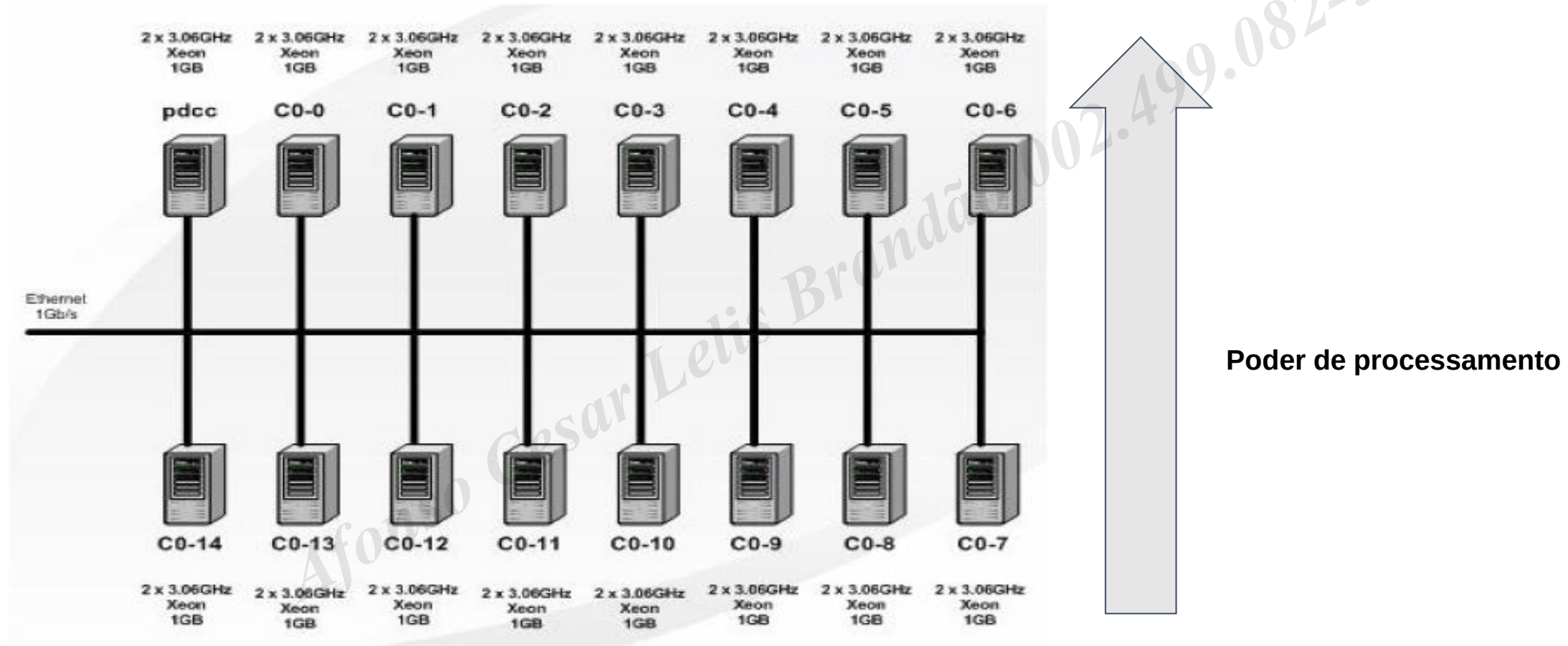
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Spark e a introdução do RDD



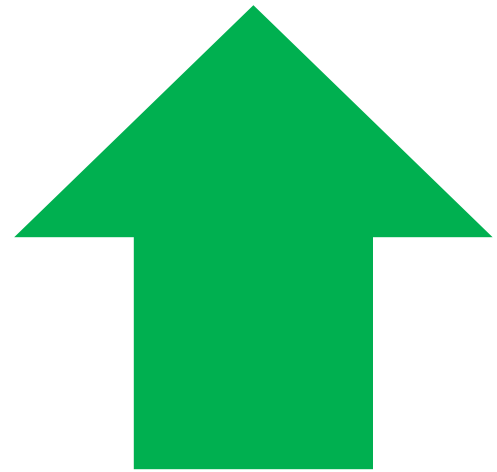
Fonte: <https://runawayhorse001.github.io/LearningApacheSpark/introduction.html>

Possibilidade de criar cluster de processamento

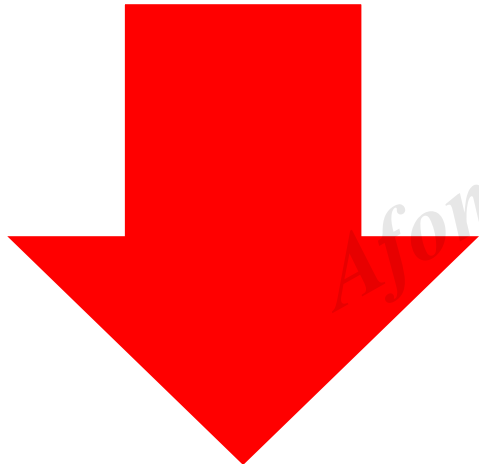


Fonte: <https://runawayhorse001.github.io/LearningApacheSpark/introduction.html>

Estratégias e boas práticas



Performance



Custos

1. Usar um framework de big data
2. Comprimir os dados
3. Particionar os dados

Pensar na compressão de dados



Parquet

... Até **87%** de redução de tamanho

... Até **34x** mais rápido para carregar os dados

... Até **99%** de redução de custos

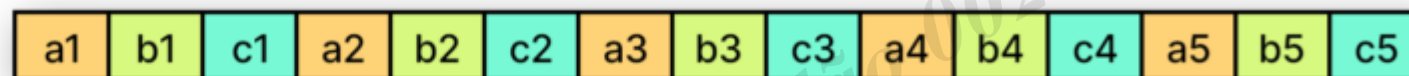
Fonte: <https://www.databricks.com/glossary/what-is-parquet>

Como ele funciona?

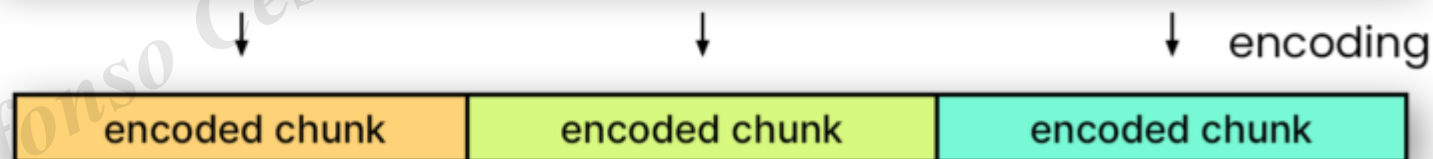
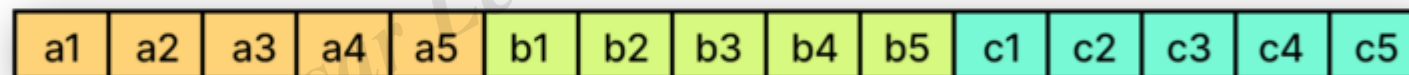
Logical table representation

a	b	c
a1	b1	c1
a2	b2	c2
a3	b3	c3
a4	b4	c4
a5	b5	c5

Row Layout



Column Layout



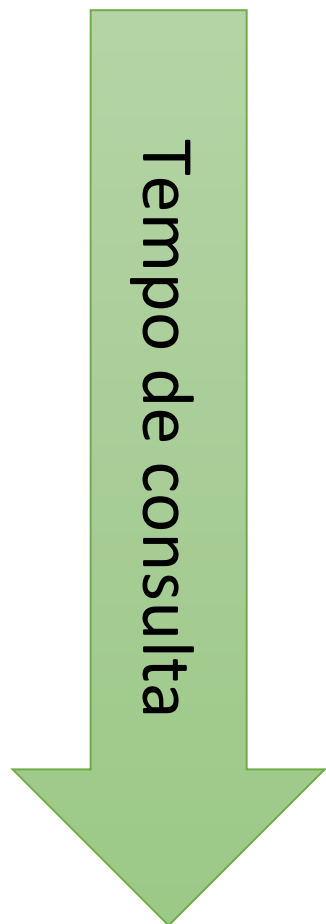
Fonte: <https://www.dremio.com/resources/guides/intro-apache-parquet/>

Comprime mesmo?

Dataset	Size on Amazon S3	Query Run time	Data Scanned	Cost
Data stored as CSV files	1 TB	236 seconds	1.15 TB	\$5.75
Data stored in Apache Parquet format*	130 GB	6.78 seconds	2.51 GB	\$0.01
Savings / Speedup	87% less with Parquet	34x faster	99% less data scanned	99.7% savings

Fonte: <https://blog.openbridge.com/how-to-be-a-hero-with-powerful-parquet-google-and-amazon-f2ae0f35ee04>

Particionar os dados



Tempo de consulta

.../tabela_de_vendas/

.../tabela_de_vendas/ano=**2022**

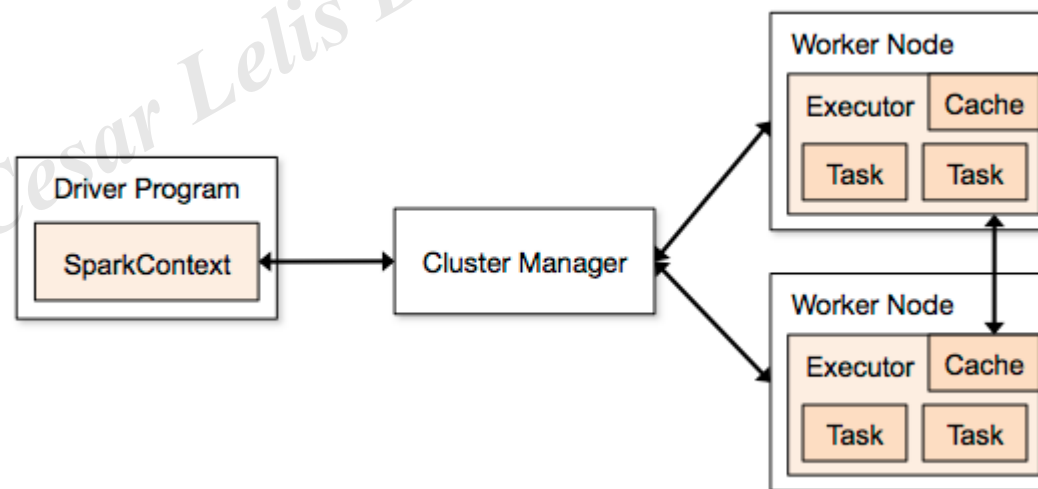
.../tabela_de_vendas/ano=**2022**/mes=**8**

Drasticamente!!!!

... E em 15 anos?

Framework que iremos utilizar

- Spark
 - Processamento distribuído em memória
 - Ferramenta open source
 - Possui biblioteca built-in de Machine Learning



Fonte: <https://spark.apache.org/docs/1.1.0/cluster-overview.html>

Mãos à obra!





OBRIGADO!

[linkedin.com/in/helderprado](https://www.linkedin.com/in/helderprado)